



Multivariable Calculus (MAT-113)

BS- Artificial Intelligence, Cyber Security and Multimedia & Gaming

General Information

Credit Hours	03(3+0)
Prerequisite	Calculus and Analytical Geometry
Course Instructor	Yasir Ali
Email address	Yasir.faculty@aror.edu.pk
Office location	Faculty offices, 2 nd Floor, B1
Consulting Hours	

Mapping of CLOs with GAs and Bloom's Taxonomy Levels

CLO	Graduate Attribute (GA)	Bloom's Taxonomy Level
CLO 1: Understand the fundamental concepts of multivariable calculus, including multivariable functions, partial derivatives, and multiple integrals.	Knowledge and Problem Solving	Remembering, Understanding
CLO 2: Apply multivariable calculus techniques to solve optimization problems, including finding maximum and minimum values of functions subject to constraints.	Critical Thinking and Analytical Skills	Applying, Analyzing
CLO 3: Evaluate line integrals, surface integrals, and volume integrals, and apply the fundamental theorems of vector calculus, such as Green's Theorem.	Critical Thinking, Problem Solving	Evaluating, Creating

Assessment

S. No	Assessment Activities	Percentage	Total Activities
1.	Sessional: Quizzes, Assignments and Presentation	30%	2,4,0
2.	Mid Term Exam	30%	1
3.	Final Exam	40%	1

Course Content

Date/ Week	Topics	Course % covered
Week 01	Lecture 01: Introduction to Multivariable Functions, Recall single variable functions Lecture 02: Functions of several variables(two & three variables, 3D graphs by GeoGebra & Desmos) Lecture 03: Domain & range of multivariable functions(Graphing the Doman of Multivariable Functions by GeoGebra & Desmos)	7.69%
Week 02	Lecture 01: Limits of Multivariable functions with examples Lecture 02: Continuity of Multivariable Functions Lecture 03: Recall Derivative of a function in one variable	15.38%
Week 03	Lecture 01: Introduction to Partial Derivatives Lecture 02: Computing Partial Derivatives Lecture 03: Higher derivatives & Clairaut's theorem	23.07%
Week 04	Lecture 01: Chain Rule (case 1 & 2) Lecture 02: implicit differentiation Lecture 03: Directional Derivatives & Gradient vector	30.76%
Week 05	Lecture 01: introduction of Maximum & Minimum values Lecture 02: Local maximum & Local Minimum (2 nd derivative test) Lecture 03: Absolute maximum & Absolute Minimum values	38.46%
Week 06	Lecture 01: Real life problems on Maximum & minimum values Lecture 02: Introduction to Multiple integral (recall single integral) Lecture 03: first lecture on Double Integrals over rectangles	46.15%
Week 07	Lecture 01: second lecture on Double Integrals over rectangles Lecture 02: third lecture on Double Integrals over rectangles	53.84%
Week 08	Lecture 01: Double Integrals over general Regions Lecture 02: Applications of Double Integrals Lecture 03: Graphical Representation of double integrals	61.53%
Week 09	Mid Term Exam From two worksheets (chapter 14 & 15)	
Week 10	Lecture 01: Introduction to Triple Integrals Lecture 02: Triple Integrals over Rectangular Boxes Lecture 03: changing the order of integration+ Applications of triple integrals	69.23%
Week 11	Lecture 01: Introduction of vector fields, scalar fields & gradient fields Lecture 02: Line Integrals in space Lecture 03: Line Integrals of Vector Fields	76.92%
Week 12	Lecture 01: The fundamental theorems for line integrals Lecture 02: conservative vector fields and potential functions Lecture 03: Problems on conservative vector fields and potential functions	84.61%
Week 13	Lecture 01: Green's Theorem Lecture 02: Applications of Green's Theorem Lecture 03: Solving problems of Green's Theorem	92.30%
Week 14	Lecture 01: introduction of curl and divergence Lecture 02: The curl Problems Lecture 03: The Divergence Problems	100%
Week 15	<u>Revision (worksheet 3 and 4 from chapter 15 & 16)</u>	

Week 16	<u><i>Final Exam</i></u>	
------------	--------------------------	--

Textbook

1. *Calculus early transcendentals* by James Stewart, Daniel clegg, Saleem Watson 9th edition.